

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE-4040 - Digital Electronics
Spring Term 2026

Lab. 4-Inverter Noise Margins

Issued: 2/11/2026

Due: 2/13/2026

Given

Below is a table of process parameters for a generic 0.13 μm technology.

0.13 μm Process technology parameters

Typical Technology Parameters	Symbol	NMOS	PMOS
Gate thickness	t_{ox} [Å]	22	22
Poly doping level	N_D [cm^{-3}]	3×10^{20}	—
Poly doping level	N_A [cm^{-3}]	—	3×10^{20}
Substrate doping level	N_A [cm^{-3}]	3×10^{17}	—
N-well doping level	N_D [cm^{-3}]	—	3×10^{17}
Number of surface state charges	N_{ss} [cm^{-2}]	6×10^{11}	6×10^{11}
Electron Mobility	μ_n [cm^2/Vs]	270	—
Hole Mobility	μ_p [cm^2/Vs]	—	70
Critical Field x Length	E_{CL} [V]	0.6	2.4
Saturation velocity	v_{sat} [cm/s]	8×10^6	8×10^6
Gamma	γ [$\text{V}^{1/2}$]	0.2	0.2
Oxide capacitance	C_{ox} [F/cm^2]	1.6×10^{-6}	1.6×10^{-6}
Threshold voltage	V_T [V]	0.4	-0.4
2 x Fermi voltage	$2 \phi_F $ [V]	0.88	0.88
Channel length modulation	λ [V^{-1}]	0.2	0.2

FOR ALL LTSPICE SIMULATION THERE IS A NEW MOS2.TXT SPICE MODEL ON LMS. THIS MODEL PRODUCES RESULTS THAT VERY CLOSELY MATCH YOUR HJS TEXTBOOK AND NOTES EQUATIONS.

Solving for V_{IL}

V_{IL} is calculated with the following equation:

$$V_{IL} = \frac{2V_{out} - V_{DD} - |V_{TH}| + (k_N/k_P)(V_{TN})}{1 + (k_N/k_P)}$$

From lecture slide 7-18

Here V_{OUT} is an unknown. For an inverter we expect V_{OUT} to be high when input is low, so we initially guess V_{OH} as the V_{OUT} value. After calculating V_{IL} , we can now solve for a more accurate V_{OUT} value with the following equation:

At $V_{in} = V_{IL}$ (Region II), $I_{DN}(sat) = I_{DP}(lin)$

$$\frac{W_N v_{sat} C_{ox} (V_{in} - V_{TN})^2}{(V_{in} - V_{TN}) + E_{CN} L_N} = \frac{W_P}{L_P} \frac{\mu_P C_{ox}}{\left(1 + \frac{V_{DD} - V_{out}}{E_{CP} L_P}\right)} \times \left[(V_{DD} - V_{in} - |V_{TH}|)(V_{DD} - V_{out}) - \frac{(V_{DD} - V_{out})^2}{2} \right]$$

From lecture notes slide 7-17

This now gives us a reduced V_{OUT} value. Recalculate V_{IL} again. If there is no significant change, then we are done. If there is, then recalculate V_{OUT} and V_{IL} repeatedly until there is no significant change.

Solving for V_{IH}

V_{IH} is calculated with the following equation:

$$V_{IH} = \frac{2V_{out} + V_{TN} + (k_P/k_N)(V_{DD} - |V_{TH}|)}{1 + (k_P/k_N)}$$

From lecture slide 7-20

Here V_{OUT} is an unknown. For an inverter we expect V_{OUT} to be low when input is high, so we initially guess V_{OL} as the V_{OUT} value. After calculating V_{IH} , we can now solve for a more accurate V_{OUT} value with the following equation:

At $V_{in} = V_{IH}$ (Region IV), $I_{DN}(lin) = I_{DP}(sat)$

$$\frac{W_N}{L_N} \frac{\mu_n C_{ox}}{\left(1 + \frac{V_{out}}{E_{CN} L_N}\right)} \left[(V_{in} - V_{TN})V_{out} - \frac{V_{out}^2}{2} \right] = \frac{W_P v_{sat} C_{ox} (V_{DD} - V_{in} - |V_{TH}|)^2}{(V_{DD} - V_{in} - |V_{TH}|) + E_{CP} L_P}$$

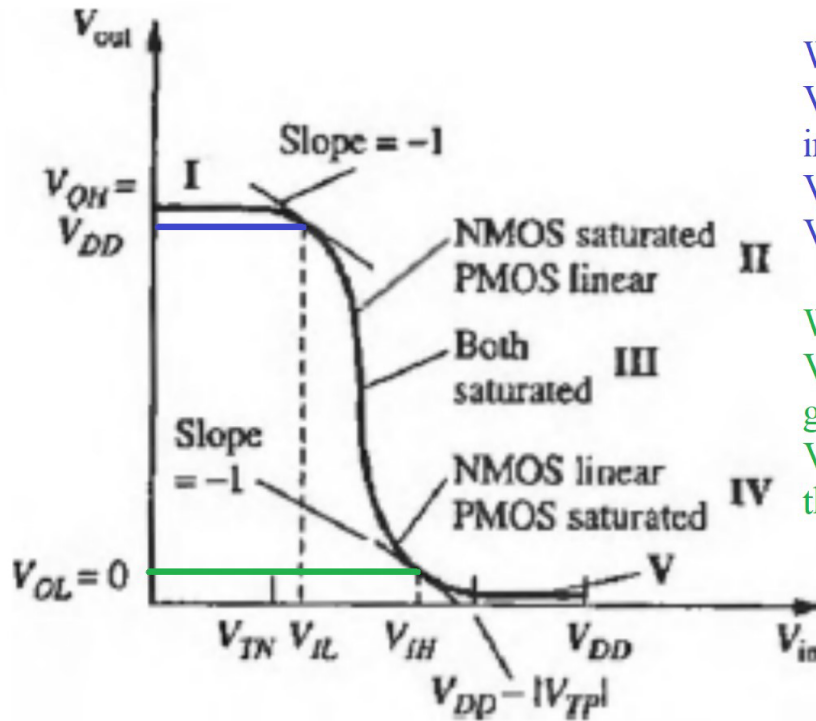
From lecture slide 7-19

This now gives us a reduced V_{OUT} value. Recalculate V_{IL} again. If there is no significant change, then we are done. If there is, then recalculate V_{OUT} and V_{IL} repeatedly until there is no significant change.

Use an equation solver aid when solving the equations above.

Inverter VTC Curve showing non-ideal V_{OUT} values for V_{IH} and V_{IL}

INV Regions



When finding V_{IL} , V_{OH} is good initial guess for V_{OUT} , but V_{OUT} at V_{IL} is less than V_{OH}

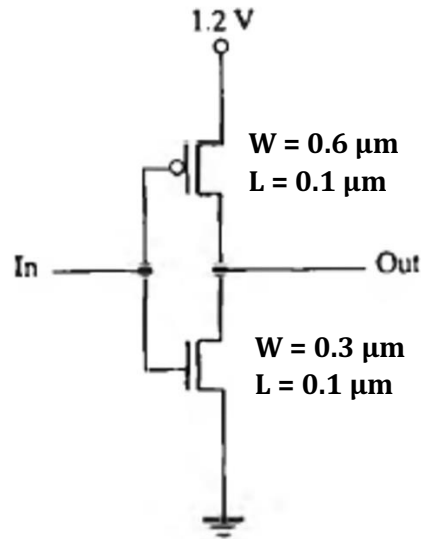
When finding V_{IH} , V_{OL} is a good initial guess for V_{OUT} , but V_{OUT} at V_{IH} is greater than V_{OL}

Question 1: Based on Problem 4.10 (D.A. Hodges, H.G. Jackson, and R.A. Saleh, *Analysis and Design of Digital Integrated Circuits*, 3rd Ed., McGraw-Hill, 2004.)

Objectives:

- Familiarize yourself with the calculation of V_{OH} , V_{OL} , V_{IL} , V_{IH} , and V_S , parameters required to calculate noise margins.

Answer the following questions for the circuit below:



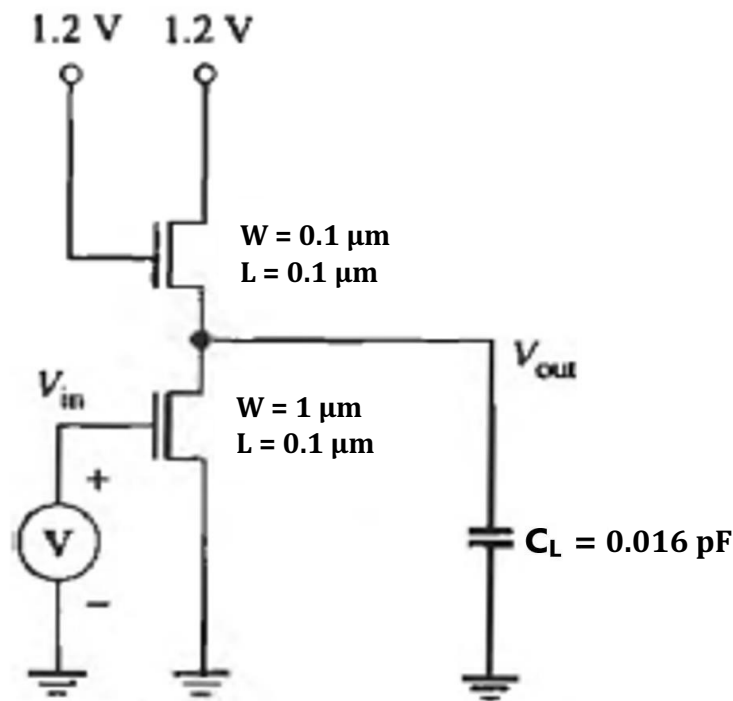
- Calculate V_{OH} , V_{OL} , V_{IL} , V_{IH} , and V_S .
- Describe in words the relative change in the part(a) voltages if the width of the PMOS transistor doubles.
- Calculate V_S again with a double width PMOS.
- Sketch the VTC (Voltage Transfer Curve) for the inverter in both PMOS width cases in one plot.

Question 2: Based on Problem 4.13 (D.A. Hodges, H.G. Jackson, and R.A. Saleh, *Analysis and Design of Digital Integrated Circuits*, 3rd Ed., McGraw-Hill, 2004.)

Objectives:

- Compare hand vs simulator results for Noise Margins.

Answer the following questions for the circuit below:



- Using LTSpice determine the noise margins by plotting the VTC.
 - Remember to set capacitor initial conditions if needed.
- Calculate V_s by hand and hand draw the VTC. Do not bother with V_{IL} , V_{IH} calculation.
 - How does this compare to the simulated VTC?

LTSpice:

- Be sure to use the new MOS2.txt MOSFET SPICE model definition file on LMS.

Report Submission:

Submit your work on LMS under **Labs**

- Show your work.
- Include images of your hand calculations with work and final results.
- Include screenshots of LTSpice with all circuit schematics with all output plots, **clearly showing all hand calculated parameters.**
- Due date: Friday at 11:59pm.